

$$\frac{\partial \phi}{\partial t} = M \nabla^2 \mu + \phi(1 - \phi) \int u \rho \, d\rho$$

$$\int u \, d\rho = \phi$$

$$\frac{\partial(u\phi)}{\partial t} = D \nabla^2(\phi u) + \phi \left(\phi(1 - \phi)u\rho + D_\rho \frac{\partial^2 u}{\partial \rho^2} \right)$$

$$\frac{\partial u}{\partial t} = -\frac{u}{\phi} \frac{\partial \phi}{\partial t} + \frac{D}{\phi} \nabla^2(\phi u) + \phi(1 - \phi)u\rho + D_\rho \frac{\partial^2 u}{\partial \rho^2}$$